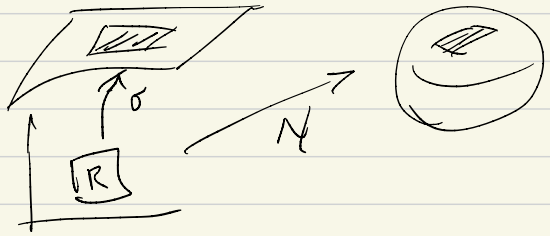
Interpretations

1) Gauss curvature = area distortion by the Gauss map



$$\text{Area}(\sigma(R)) = \int_R \|\sigma_1 \times \sigma_2\| \, du^1 du^2$$

$$\text{Area}(N(R)) = \int_R \|N_1 \times N_2\| \, du^1 du^2$$

$$N_i = -W_i^j \sigma_j \implies N_1 \times N_2 = (\det W) \cdot \sigma_1 \times \sigma_2$$

$$= K \, \sigma_1 \times \sigma_2$$

$$|K(p)| = \lim_{R \to \{p\}} \frac{\text{Area}(N(R))}{\text{Area}(\sigma(R))}$$

sign of K depends on whether N preserves or reverses orientation

In this interpretation, curvature measures the 2-dim movement of the normal vector

2) Principal curvatures

Recall normal curvature at $\gamma : I \to S$ w.r.t. N

$$K_n(p, \dot\gamma) = N \cdot \ddot\gamma = \text{II}(\dot\gamma, \dot\gamma) \quad \text{if } \|\dot\gamma\| = 1$$

define $$K_+(p) = \max_{\|X\|=1} \text{II}(X, X)$$

$$K_-(p) = \min_{\|X\|=1} \text{II}(X, X)$$

Since $\underline{\mathrm{II}}(\cdot,\cdot) = g(-dN(\cdot),\cdot)$, and $\underline{\mathrm{II}}$ is symmetric,

$-dN$ is self-adjoint w.r.t. $g(\cdot,\cdot)$

Min-max principle $\Longrightarrow$ $\kappa_\pm$ are the eigenvalues of $-dN_p$

$\therefore K = \kappa_+ \kappa_-$ and $H = \frac{1}{2}(\kappa_+ + \kappa_-)$

3) mean curvature: let $\sigma : U \to S$ be a regular patch and $R \subset U$ a cpt. region

$$\mathrm{Area}(\sigma(R)) = \int_R \|\sigma_1 \times \sigma_2\| \, du^1 du^2$$

Perturbation: $\sigma_\varepsilon = \sigma + \varepsilon N\phi$ where ϕ is a smooth bump-fn. in R

Exercise: $\mathrm{Area}(\sigma_\varepsilon(R)) = \mathrm{Area}(\sigma(R)) - 2\varepsilon \int_R \phi H \, dA + O(\varepsilon^2)$

$H = 0 \iff \mathrm{Area}(\sigma_\varepsilon(R))$ has a critical pt. at $\varepsilon = 0$ for all ϕ

S has minimal area (under local perturbations) $\iff H = 0$

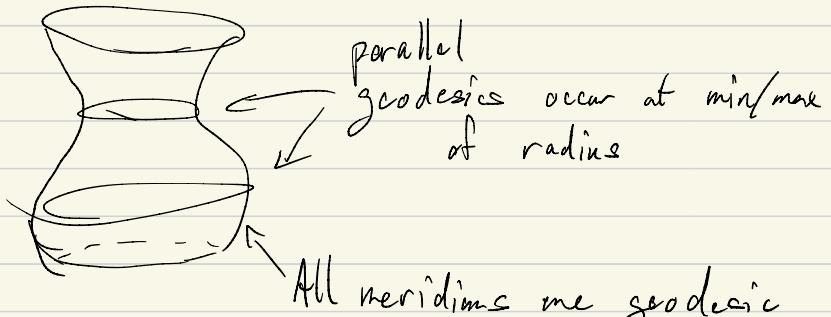
Geodesics

define: A curve $\gamma: I \to S$ is a <u>geodesic</u> if

$$k_g = 0$$

For a unit speed curve, geodesic $\iff \ddot{\gamma} \parallel N$

example: surface of revolution



Note: $\ddot{\gamma} \parallel N \implies \ddot{\gamma} \cdot \dot{\gamma} = 0 \implies \frac{d}{dt}\|\dot{\gamma}\| = 0$

(constant speed)

<u>local equations</u>:

Write $\gamma = (\gamma^1, \gamma^2)$ in a coord. patch $\sigma(u^1, u^2)$

constant speed geodesic $\iff \ddot{\gamma} \perp \sigma_1$ and σ_2

Consider $\frac{d}{dt}(\dot{\gamma} \cdot \sigma_i) = \ddot{\gamma} \cdot \sigma_i + \dot{\gamma} \cdot \dot{\sigma}_i$

γ const. speed geodesic $\iff \frac{d}{dt}(\dot{\gamma} \cdot \sigma_i) = \dot{\gamma} \cdot \dot{\sigma}_i$ for $i = 1, 2$

In the basis $\dot{\gamma} = \dot{\gamma}^j \sigma_j$, $\dot{\sigma}_i = \dot{\gamma}^k \sigma_{ik}$

The equation becomes

$$\frac{d}{dt}\left(\dot{\gamma}^j \sigma_j \cdot \sigma_i\right) = \left(\dot{\gamma}\sigma_j\right) \cdot \left(\dot{\gamma}^k \sigma_{ik}\right)$$

By def. $\sigma_j \cdot \sigma_i = g_{ij}$

$$\partial_i g_{jk} = \sigma_{ji} \cdot \sigma_k + \sigma_j \cdot \sigma_{ki}$$

$$\therefore \quad \ddot{\gamma}^j \dot{\gamma}^k (\partial_i g_{jk}) = 2 \ddot{\gamma}^j \dot{\gamma}^k \sigma_j \cdot \sigma_{ik}$$

<u>Lemma</u> γ is a constant speed geodesic if and only if

$$\boxed{\frac{d}{dt}(g_{ij}\dot{\gamma}^j) = \frac{1}{2}\ddot{\gamma}^j \dot{\gamma}^k (\partial_i g_{jk})} \quad \text{for } i=1,2$$

(geodesic property is intrinsic, depends only on g